Name: _____

WORKSHEET

Applications of the dot product

1 Calculate the work done in each of the following situations.

a



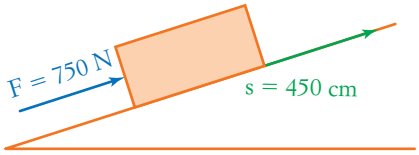
b



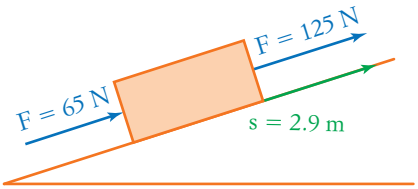
c



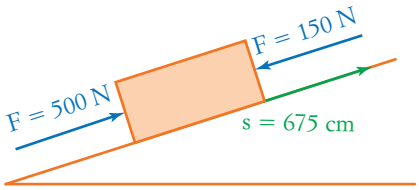
d



e



f



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2 Calculate the work done in each of the following situations.

a A horizontal force of 50 N that moves an object 40 m.

b A horizontal force of 1000 N that moves an object 225 cm.

c An angled force of 125 N that moves an object 7.5 m in the same direction.

d Horizontal forces of 275 N and −105 N that move an object 20 m.

e Horizontal forces of 500 N and −50 N that move an object 115 cm.

f Angled forces of 200 N and −150 N that move an object 87 cm in the same direction.

3 For the following situations:

i Draw a sketch of the situation

ii Find the vectors **F** and **s**

iii Find the energy gained by the mass/work done by the force.

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a A mass of 16 kg accelerates 42 m down a slope at 15° to the horizontal

b A mass of 75 kg accelerates 2.1 km down a slope at 7° to the horizontal

c A 21 kg object falls 35 m freely under the influence of gravity.

d An object is subjected to an upwards force of 250 N and a force of 500 N at an angle of 45° to the horizontal, moving 10 m.

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e A force of 40 N acts on an object at an angle of 35° from the horizontal in addition to a force of 50 N acting on an angle of 15° from the horizontal, moving the object 4 m horizontally.

f A force of 100 N acts on an object at an angle of 25° from the horizontal in addition to a force of 75 N acting on an angle of 65° from the vertical, moving the object 1.1 km horizontally.

4 An object undergoes a displacement of $\mathbf{s} = 5\mathbf{i} + 2\mathbf{j}$ as a result of a force $\mathbf{F}$ newtons. Calculate the work done in each of the following cases.

a $\mathbf{F} = 3\mathbf{i} - 5\mathbf{j}$

b $\mathbf{F} = -2\mathbf{i} - 4\mathbf{j}$

c $\mathbf{F} = 4\mathbf{i} + 7\mathbf{j}$

d $\mathbf{F} = -7\mathbf{i} + \mathbf{j}$

e $\mathbf{F} = \mathbf{i} - 5\mathbf{j}$

f $\mathbf{F} = 11\mathbf{i} + 6\mathbf{j}$

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Answers

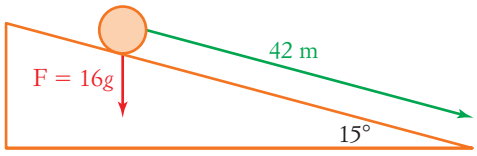
1 **a** 600 J **c** 900 J **e** 551 J

b 3150 J **d** 3375 J **f** 2362.5 J

2 **a** 2000 J **c** 937.5 J **e** 517.5 J

b 2250 J **d** 3400 J **f** 43.5 J

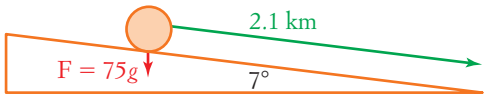
3 **a** **i**



ii $\mathbf{F} = 156.96$ N vertically, $\mathbf{s} = 42$ m at 75° to the vertical $= 42 \cos(75°)$

iii $\mathbf{W} = 1706.2$ J

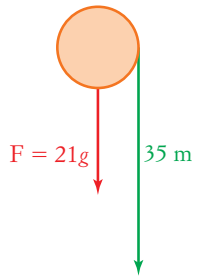
b **i**



ii $\mathbf{F} = 735.75$ N, $\mathbf{s} = 2100$ m at 83° to the vertical $= 2100 \cos(83°)$

iii $\mathbf{W} = 188\,297.3$ J

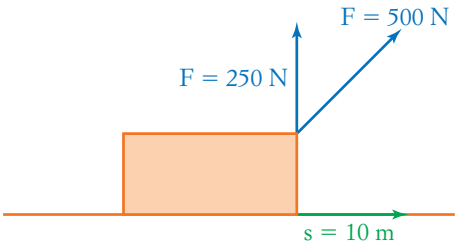
c **i**



ii $\mathbf{F} = 206.01$ N, $\mathbf{s} = 35$

iii $\mathbf{W} = 7210.35$ J

d **i**

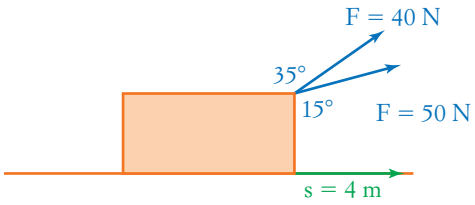


ii $\mathbf{F} = 433$ N, $\mathbf{s} = 10$ m

iii $\mathbf{W} = 4330$ J

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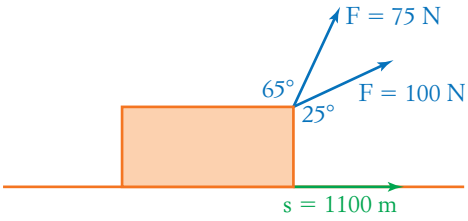
e i



ii $\mathbf{F} = 40\cos(35°) + 50\cos(15°) = 81.06$ N, $\mathbf{s} = 4$

iii $\mathbf{W} = 324.25$ J

f i



ii $\mathbf{F} = 75\cos(65) + 100\cos(25) = 122.33$, $\mathbf{s} = 1100$

iii $\mathbf{W} = 134\,559.86$ J

4 a 5 J

b −18 J

c 34 J

d −33 J

e −5 J

f 67 J

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